Leon Sandler
Independent Researcher
Northbrook, Illinois, USA
sandler.leon@gmail.com
ORCID 0009-0007-4584-808X

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The Editor-in-Chief
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Dear Editor,

I submit the manuscript “Thermal Shielding, Melt Kinetics and Surge Control of a Steam-Assisted Pipeline Cleaning Crawler: A Monte Carlo Study and Its Deployment Limits” as a research article for consideration in Applied Thermal Engineering.

Steam-assisted tools are an attractive alternative to abrasive cleaning of hydrocarbon pipelines, but a tool that works in saturated steam near 200 °C must keep its battery and electronics cool, melt the deposit fast enough to matter, and control its speed when a blockage releases. The manuscript treats these three problems for one concept, a motorized, tracked crawler with a sealing skirt, with a probabilistic reduced-order model: transient conduction through an aluminium, silica-aerogel and titanium wall coupled to a lumped battery pack, Nusselt film condensation on the deposit, a blockage-release transient with an orifice bypass, and a steam budget. It then tests the concept against the operating envelope of a published industrial scouting challenge (pipes of 4 to 16 inches and up to 1 km, no fixed launcher, access through a connection aligned with the flow).

The manuscript makes four contributions. (i) A thermal result with its condition: with 40 mm of aerogel the pack stays at or below 55 °C for 45 min in 98.9% of 3,000 draws, but the probability falls to 84% at 35 °C and 22% at 45 °C, and the hull does not fit pipes below 8 in. (ii) A finding about melt kinetics: the cleaned range of 138 m per deployment that follows from instant removal of the molten deposit falls to 27 m with a retained film 0.2 mm thick and to 2 m if none drains, so a quantity nobody has measured, the retained melt film, decides the reach. (iii) A finding about logistics: no 45 min deployment reaches 300 m, a 1 km line needs a median of 8 deployments, and heating the pipe wall adds 1.8 times the melting steam. (iv) A finding about surge control: a bypass valve that is closed at the stall cannot arrest the first surge peak, because the tool reaches 1 m/s in about 6 ms, a time scale that the earlier force-factor treatment of the relief hid; a bypass that is open during the stall is the design alternative.

The study is explicit about what it is. Every input distribution is an assumed design-stage value, because no published distribution of these deposits, steam conditions or tool was found; the model is verified numerically (the first suite is retained unchanged, every added mechanism reduces to it when switched off, and 27 automated checks pass) but not validated physically; and the conclusions rank the measurements that would matter, rather than predicting performance. The concept is at technology readiness level 2 to 3, and the manuscript says so.

The code and results are openly available at https://github.com/sandlerleon/tamprc-pipeline-crawler-mc and archived at https://doi.org/10.5281/zenodo.24045599 (code, all versions); the manuscript and its Supplementary Information are archived at https://doi.org/10.5281/zenodo.24045595 (all versions). One command regenerates every number and figure.

Disclosure. The concept was prepared as a response to a public scouting challenge on a third-party innovation platform. I have no financial relationship with the seeker or the platform, and the work received no external funding.

Author affiliation and email address. I am an independent researcher with no current institutional affiliation, and I am the sole author. The corresponding-author email address, sandler.leon@gmail.com, is my personal address.

A generative AI tool (Claude, Anthropic) assisted with code development, numerical experiments, drafting and editing; I reviewed and checked all of it and take full responsibility for the content, as disclosed in the manuscript. The manuscript is original, has not been published elsewhere and is not under consideration at any other journal. I am the sole author and have no competing interests.

Thank you for considering this submission.

Sincerely,

Leon Sandler